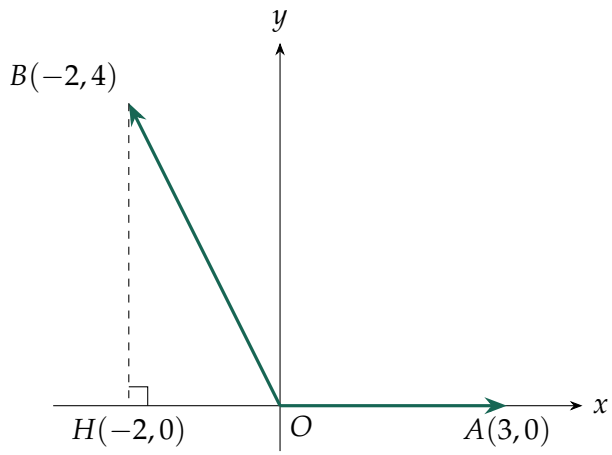
## 3.1 Projection and angle

**Key point · Definition and results.** For two nonzero vectors $\vec{u}, \vec{v}$, their scalar product is the real number $\vec{u} \cdot \vec{v} = \|\vec{u}\| \, \|\vec{v}\| \cos\theta$, where $\theta \in [0, \pi]$ is their unoriented angle. With a zero vector, the product is 0. If $H$ is the projection of B onto $(OA)$, $\overrightarrow{OA} \cdot \overrightarrow{OB}$ equals $OA \times OH$ when H is on the ray $[OA)$, and its opposite otherwise.

**Status and justification.** Definition; the equivalence with the projection follows from the cosine in right triangles, with a negative signed projection for an obtuse angle.

**Example 1.** For two vectors of norms 3 and 4 forming 60°, the product is $\dfrac{3 \times 4 \times 1}{2} = 6$. With 120°, it is $-6$. If $A(3,0)$ and $B(-2,4)$, the projection is $H(-2,0)$ and the product is $-6$.




The projection H lies on the negative side of the horizontal axis. Hence $\overrightarrow{OA} \cdot \overrightarrow{OB} = 3 \times (-2) = -6$.

**Watch out.** The result is a number, not a vector. A right angle corresponds to 0, not to 1.

**Application.** Two vectors have norms 2 and 5 and form 90°, then 180°. Calculate the products.

## 3.2 Coordinates, norm and orthogonality

**Key point · Definition and results.** In an orthonormal basis, for $\vec{u} = (x, y)$ and $\vec{v} = (x', y')$, $\vec{u} \cdot \vec{v} = xx' + yy'$ and $\|\vec{u}\|^2 = x^2 + y^2$. The vectors are orthogonal if and only if their product is zero. For an orthonormal basis $(\vec{i}, \vec{j})$, $x = \vec{u} \cdot \vec{i}$ and $y = \vec{u} \cdot \vec{j}$.

**Status and justification.** Properties proved using bilinearity: expand $(x\vec{i} + y\vec{j}) \cdot (x'\vec{i} + y'\vec{j})$ and use $\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = 1$, $\vec{i} \cdot \vec{j} = 0$.

**Example 2.** $\vec{u} = (2,3)$, $\vec{v} = (3,-2)$: product $6 - 6 = 0$. The two vectors are orthogonal. Their norm is $\sqrt{13}$. The product of $(2,3)$ with $(1,0)$ is 2: we recover the x-coordinate.

**Watch out.** The usual formula assumes an orthonormal basis. Collinearity and orthogonality are different properties.

**Application.** For which real number t are the vectors $(t, 2)$ and $(3, -6)$ orthogonal?

## 3.3 Vector operations and identities

**Key point · Definition and results.** The scalar product is symmetric and bilinear: $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$ and $(\lambda \vec{u}) \cdot \vec{v} = \lambda(\vec{u} \cdot \vec{v})$. Hence $\|\vec{u} \pm \vec{v}\|^2 = \|\vec{u}\|^2 \pm 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2$. It follows that $\vec{u} \cdot \vec{v} = \dfrac{\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2}{2}$.

**Status and justification.** Bilinearity accepted from the geometric definition; symmetry comes from the cosine. The identities are proved by expanding $(\vec{u} \pm \vec{v}) \cdot (\vec{u} \pm \vec{v})$.

**Example 3.** If $\|\vec{u}\| = 3$, $\|\vec{v}\| = 4$ and $\|\vec{u} + \vec{v}\| = 5$, the product is $\dfrac{25 - 9 - 16}{2} = 0$. We deduce orthogonality without coordinates.

**Watch out.** In general, $\|\vec{u} + \vec{v}\| \neq \|\vec{u}\| + \|\vec{v}\|$. The identities concern the squares of the norms.

**Application.** Given $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$.

## 3.4 Al-Kashi and choosing a method

**Key point · Definition and results.** In a triangle ABC, $BC^2 = AB^2 + AC^2 - 2AB\,AC\cos\widehat{BAC}$. This formula allows us to calculate the third side from two sides and the included angle, or to find the cosine of an angle from the three sides.

**Status and justification.** By Chasles, $\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB}$. Expanding its scalar square gives $BC^2 = AC^2 + AB^2 - 2\overrightarrow{AB} \cdot \overrightarrow{AC}$; then introduce the expression with the cosine.

**Example 4.** $AB = 5$, $AC = 3$ and $\widehat{BAC} = 60°$: $BC^2 = 25 + 9 - 15 = 19$, so $BC = \sqrt{19}$. With three sides, isolate the cosine before approximating the angle on a calculator, if one is allowed.

**Watch out.** The angle is included between the two sides of the product term. The minus sign is essential.

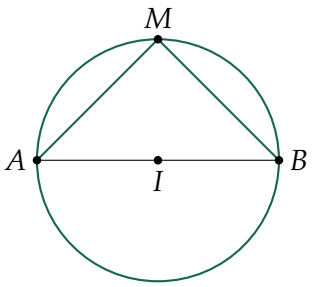
**Application.** A triangle has $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine the angle A.

## 3.5 The circle with diameter AB

**Key point · Definition and results.** For two distinct points A and B with midpoint I, $\overrightarrow{MA} \cdot \overrightarrow{MB} = MI^2 - \dfrac{AB^2}{4}$. The set of points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 0$ is the circle with centre I and radius $\dfrac{AB}{2}$, including the points A and B.

**Status and justification.** Write $\overrightarrow{MA} = \overrightarrow{MI} + \overrightarrow{IA}$ and $\overrightarrow{MB} = \overrightarrow{MI} - \overrightarrow{IA}$. The cross terms cancel; what remains is $MI^2 - IA^2$. Since $IA = \dfrac{AB}{2}$, equality to zero is equivalent to the metric equation of the circle.

**Example 5.** $A(-2,0)$ and $B(2,0)$: I is the origin. The locus is $x^2 + y^2 = 4$. For $M(0,1)$, the product is $1 - 4 = -3$; for $M(0,2)$, it is 0.




Circle with diameter AB: the vectors $\overrightarrow{MA}$ and $\overrightarrow{MB}$ are orthogonal.

**Watch out.** The endpoints belong to the locus even though "right triangle" does not apply to a degenerate triangle.

**Application.** For $A(0,0)$, $B(6,0)$, determine the points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$.